RevShop

Application Architecture & Entity Relationship Diagram

Application Architecture — MVC Monolithic

Buyer
Browses, shops, orders

Seller
Lists products, manages orders

FRONTEND LAYER — THYMELEAF TEMPLATES

index.html · login.html · register.html · cart.html · orders.html · product-detail.html · dashboard.html · favorites.html · notifications.html · CSS / JS

CONTROLLER LAYER — SPRING MVC REST CONTROLLERS

UserController · ProductController · CartController · OrderController · CategoryController · ReviewController · FavoriteController · NotificationController · SellerController · ImageController

DTO LAYER — DATA TRANSFER OBJECTS (REQUEST & RESPONSE)

UserRegistrationDTO · UserResponseDTO · ProductRequestDTO · ProductResponseDTO · OrderRequestDTO · OrderResponseDTO · CartResponseDTO

SERVICE LAYER — BUSINESS LOGIC (INTERFACE + IMPLEMENTATION)

UserService · ProductService · CartService · OrderService · ReviewService · FavoriteService · NotificationService · CategoryService · SellerService

REPOSITORY LAYER — SPRING DATA JPA

UserRepository · ProductRepository · CartRepository · OrderRepository · ReviewRepository · FavoriteRepository · NotificationRepository · PaymentRepository

DATABASE — ORACLE XE (PORT 1521) | JPA / HIBERNATE ORM

USERS · PRODUCTS · ORDERS · ORDER_ITEMS · CART · CART_ITEMS · REVIEWS · PAYMENTS · FAVORITES · NOTIFICATIONS

CROSS-CUTTING CONCERNS

Spring Security	Log4J2 Logging	Exception Handling	JUnit 5 + Mockito	Maven
BCrypt · Role Auth · CSRF	log4j2-spring.xml	GlobalExceptionHandler	9 Test Classes · 56 Tests	Dependency Management